Simple Linear Regression

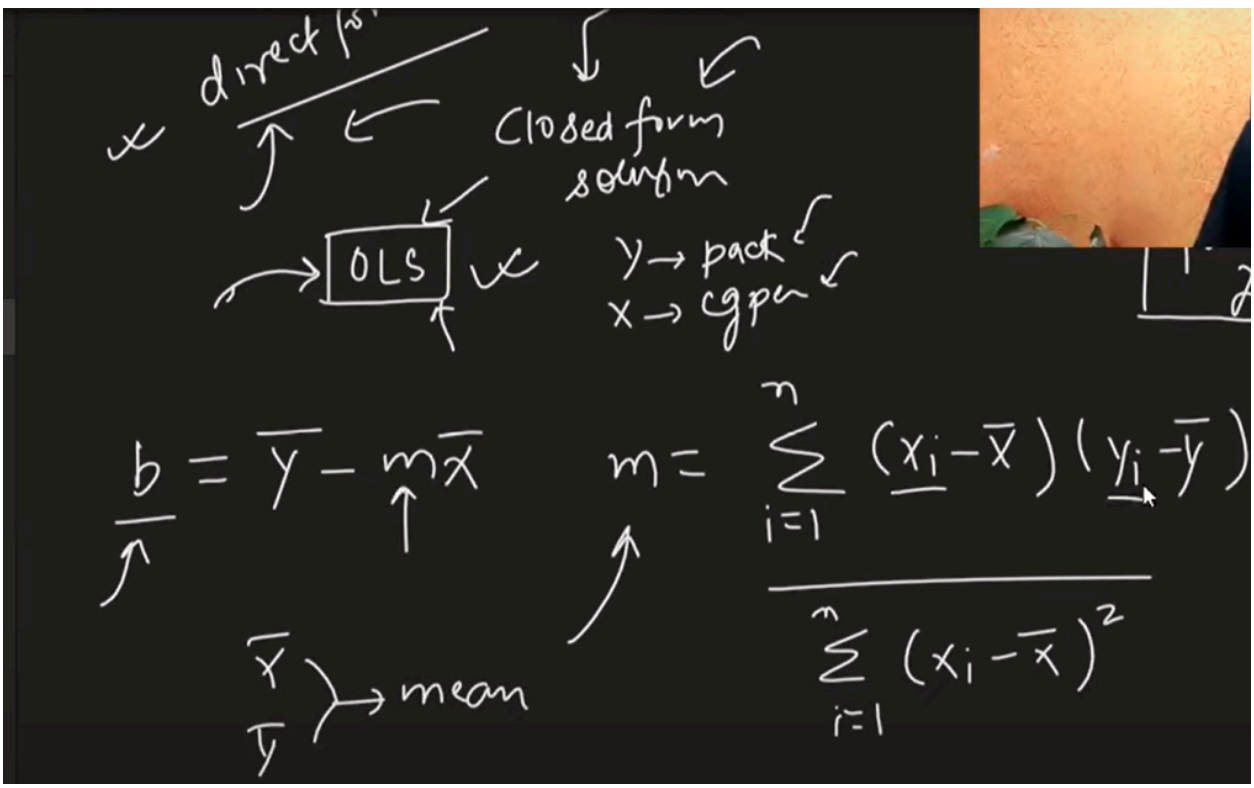
Using ScikitLearn

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

df= pd.read_csv('placement.csv')

plt.scatter(df['cgpa'], df['package'])
plt.xlabel('cgpa')
plt.ylabel('package')
plt.title('cgpa vs package')
plt.show()


from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, tes
t_size=0.2, random_state=2)

from sklearn.linear_model import LinearRegression
lr = LinearRegression()
lr.fit(X_train, Y_train)

X_test
Y_test

lr.predict(X_test.iloc[8].values.reshape(1,1))

plt.scatter(df['cgpa'], df['package'])
plt.plot(X_train, lr.predict(X_train), color='red')
plt.xlabel('cgpa')
plt.ylabel('package')
```

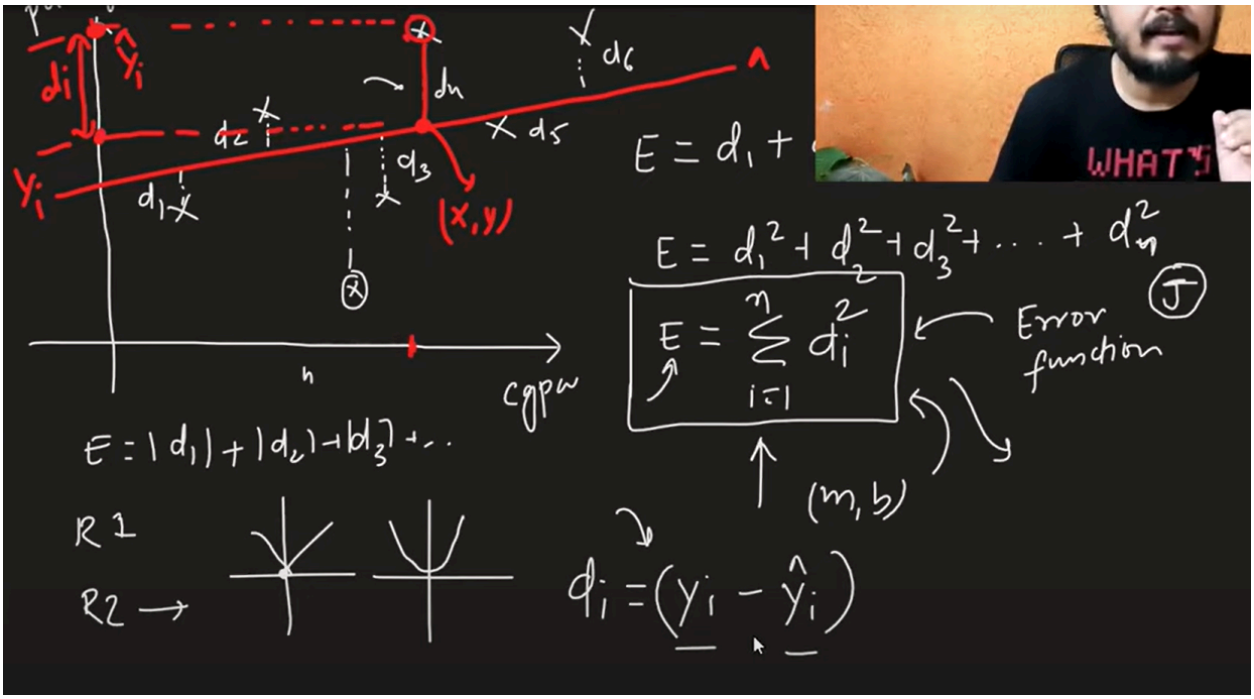
```
plt.title('cgpa vs package')
plt.show()

dhal= lr.coef_
dhal

interceptor = lr.intercept_
interceptor
```

Raw Code : without using Scikit Learn

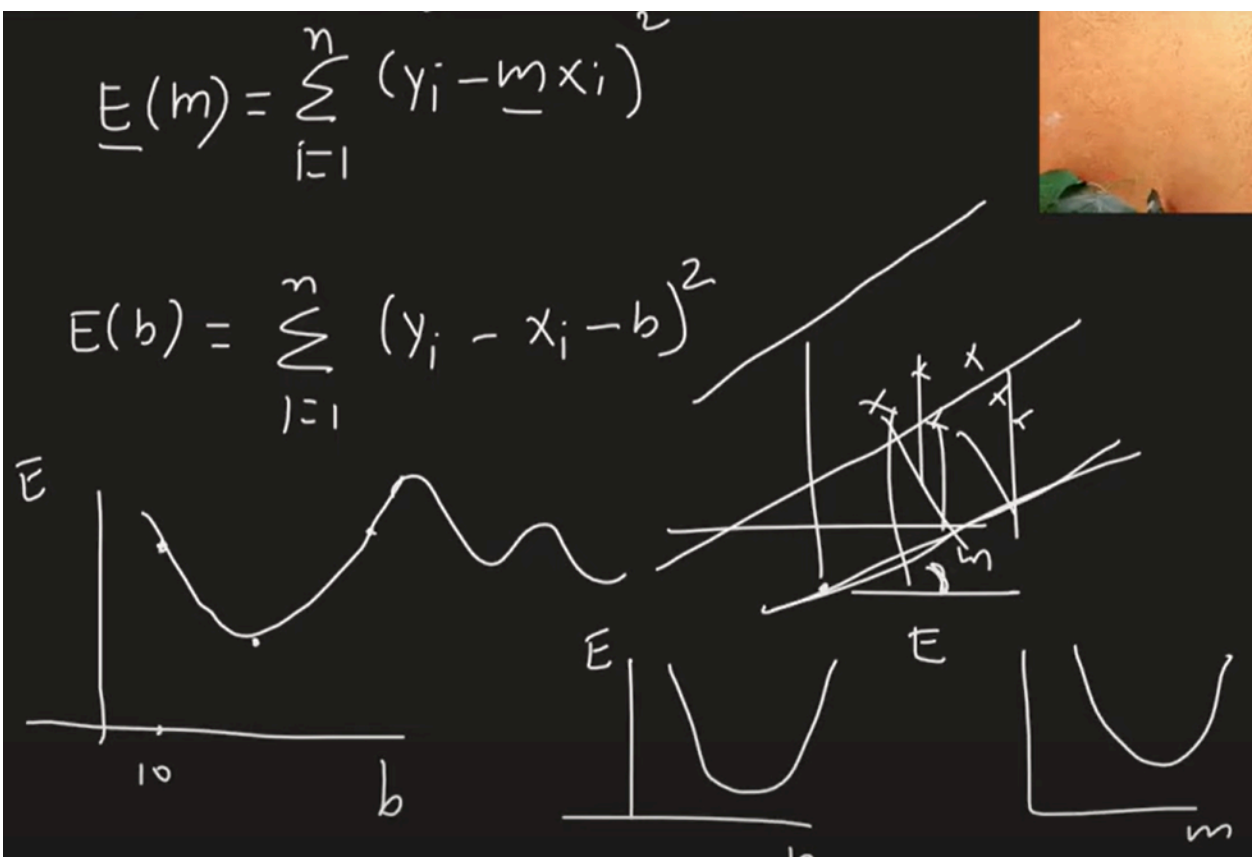
OLS : Formulas for b and m, b interceptor and m dhal



Error function formula: y^ = mx + b



Taking value of b=0, for E(m) partial diff, same for b but m=1 then



proof of b

$$\frac{\partial E}{\partial b} = \frac{\partial}{\partial b} \sum_{i=1}^{n} (y_i - mx_i - b)^2 = 0$$

$$= \sum \frac{\partial}{\partial b} (y_i - mx_i - b)^2 = 0$$

$$\Rightarrow \sum -2(y_i - mx_i - b) = 0$$

$$\Rightarrow \sum (y_i - mx_i - b) = 0$$

$$\underbrace{b + b + b + b + \ldots + b}_{n \text{ times}} = nb$$

$$\frac{\sum_{i=1}^{n} y_i}{n} \qquad (m, b) \quad n$$

$$\frac{\sum y_i}{n} - \frac{\sum mx_i}{n} - \frac{\sum b}{n} = \frac{0}{n}$$

$$\bar{y} - m\bar{x} - \frac{\cancel{n}b}{\cancel{n}} = 0$$

$$\bar{y} - m\bar{x} = b$$

$$\boxed{b = \bar{y} - m\bar{x}}$$

Proof of m

$$E = \sum (y_i - mx_i - \bar{y} + m\bar{x})^2$$

$$\frac{\partial E}{\partial m} = \sum \frac{\partial}{\partial m} (y_i - mx_i - \bar{y} + m\bar{x})^2 = 0$$

$$\Rightarrow \sum 2(y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) = 0$$

$$= \sum -2(y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x})$$

$$\Rightarrow \sum 2\left(y_i - mx_i - \bar{y} + m\bar{x}\right)\left(-\underline{x_i + \bar{x}}\right) = 0$$

$$= \sum -2\left(y_i - mx_i - \bar{y} + m\bar{x}\right)\left(x_i - \bar{x}\right) = 0$$

$$= \sum \left(\underline{y_i - mx_i - \bar{y} + m\bar{x}}\right)\left(x_i - \bar{x}\right) = 0$$

$$= \sum \left[\left(y_i - \bar{y}\right) - m\left(x_i - \bar{x}\right)\right]\left(x_i - \bar{x}\right) = 0$$

$$= \sum \left[\left(y_i - \bar{y}\right)\left(x_i - \bar{x}\right) - m\left(x_i - \bar{x}\right)^2\right] = 0$$

$$= \sum \left(y_i - \bar{y}\right)\left(x_i - \bar{x}\right) = m \sum \left(x_i - \bar{x}\right)^2$$

$$\boxed{m = \frac{\sum_{i=1}^{n} \left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right)}{\sum_{i=1}^{n} \left(x_i - \bar{x}\right)^2}}$$

Raw code of the regression using formulas :

```
class MeraLR:

def __init__(self):
        self.m=Noneself.b=None

def fit(self,X_train,y_train):

        num= 0
        den= 0

for iin range(X_train.shape[0]):

            num= num+ ((X_train[i]- X_train.mean())*(y_train
[i]- y_train.mean()))
            den= den+ ((X_train[i]- X_train.mean())*(X_train
[i]- X_train.mean()))

        self.m= num/den
        self.b= y_train.mean()- (self.m* X_train.mean())
        print(self.m)
        print(self.b)

def predict(self,X_test):

        print(X_test)

return self.m* X_test+ self.b
```

```
import numpy as np
import pandas as pd
```

```
df = pd . read_csv('placement.csv')
```

```
X= df.iloc[:,0].values
y= df.iloc[:,1].values
```

```
from sklearn.model_selectionimport train_test_split
X_train,X_test,y_train,y_test= train_test_split(X,y,test_size
=0.2,random_state=2)
lr = MeraLR()
lr.fit(X_train,y_train)
print(lr.predict(X_test[0]))
```

Regression Metrics | MSE, MAE & RMSE | R2 Score & Adjusted R2 Score

https://youtu.be/Ti7c-Hz7GSM?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH